Benemérita Universidad Autónoma de Puebla
Facultad de Ciencias de la Computación

Materia: Intercomunicación y Seguridad en Redes

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Práctica 3: Uso de Comandos DNS en Linux

Objetivo general

Que el alumno identifique, utilice y compare diferentes comandos para realizar consultas y diagnósticos relacionados con DNS en un sistema operativo Linux.

Objetivos específicos

1. Ejecutar consultas DNS usando distintos comandos (dig, host, nslookup).

2. Obtener y analizar información WHOIS de un dominio.

3. Verificar conectividad y trazar rutas hacia un dominio.

4. Analizar tráfico DNS con herramientas de captura.

5. Interpretar resultados para diagnóstico de problemas de red.

Requerimientos previos

- Equipo con Linux instalado (Ubuntu, Debian, Fedora o similar).
- Acceso a internet.
- Herramientas instaladas: dnsutils (para dig y nslookup), whois, tcpdump, nmap.
- Permisos de administrador para ejecutar tcpdump (opcional).

Procedimiento

Parte 1 – Consultas básicas DNS

Realiza los siguientes ejercicios con **dos** dominios de ejemplo elegidos por ti.

1. Consulta con dig:

```
dig www.ejemplo.com
```

- Identifica la dirección IP del dominio.
- Observa y describe el campo ANSWER SECTION.

```
migue31@cliente:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31376
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             234     IN      A       142.250.113.113
google.com.             234     IN      A       142.250.113.100
google.com.             234     IN      A       142.250.113.139
google.com.             234     IN      A       142.250.113.138
google.com.             234     IN      A       142.250.113.102
google.com.             234     IN      A       142.250.113.101

;; Query time: 19 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Jan 16 16:39:53 UTC 2026
;; MSG SIZE  rcvd: 135
```

```
migue31@cliente:~$ dig facebook.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> facebook.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 53869
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;facebook.com.                  IN      A

;; ANSWER SECTION:
facebook.com.           49      IN      A       157.240.25.35

;; Query time: 14 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Jan 16 17:30:26 UTC 2026
;; MSG SIZE  rcvd: 57

migue31@cliente:~$ _
```

dig te da información técnica detallada, incluyendo tiempos de consulta y servidores consultados.

2. Consulta específica de registros MX:

```
dig ejemplo.com MX
```

- Registra los servidores de correo que aparecen.

```
migue31@cliente:~$ dig google.com MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> google.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39440
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 10

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      MX

;; ANSWER SECTION:
google.com.             205     IN      MX      10 smtp.google.com.

;; ADDITIONAL SECTION:
smtp.google.com.        270     IN      A       172.253.135.27
smtp.google.com.        270     IN      A       142.251.186.26
smtp.google.com.        270     IN      A       142.251.186.27
smtp.google.com.        270     IN      A       142.250.115.27
smtp.google.com.        270     IN      A       142.250.115.26
smtp.google.com.        193     IN      AAAA    2607:f8b0:4023:1015::1a
smtp.google.com.        193     IN      AAAA    2607:f8b0:4023:100b::1a
smtp.google.com.        193     IN      AAAA    2607:f8b0:4023:100b::1b
smtp.google.com.        193     IN      AAAA    2607:f8b0:4023:1004::1b

;; Query time: 33 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Jan 16 16:40:54 UTC 2026
;; MSG SIZE  rcvd: 252
```

Smtp.google.com es el MX

```
migue31@cliente:~$ dig facebook.com MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> facebook.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2749
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;facebook.com.                  IN      MX

;; ANSWER SECTION:
facebook.com.           276     IN      MX      10 smtpin.vvv.facebook.com.

;; Query time: 10 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Jan 16 17:31:30 UTC 2026
;; MSG SIZE  rcvd: 68

migue31@cliente:~$ _
```

3. Consulta con host:

```
host www.ejemplo.com
```

- Compara el resultado con el de dig.

```
;; MSG SIZE  rcvd: 232

migue31@cliente:~$ host google.com
google.com has address 142.250.113.138
google.com has address 142.250.113.113
google.com has address 142.250.113.102
google.com has address 142.250.113.100
google.com has address 142.250.113.101
google.com has address 142.250.113.139
google.com has IPv6 address 2607:f8b0:4023:1000::71
google.com has IPv6 address 2607:f8b0:4023:1000::64
google.com has IPv6 address 2607:f8b0:4023:1000::66
google.com has IPv6 address 2607:f8b0:4023:1000::8a
google.com mail is handled by 10 smtp.google.com.
migue31@cliente:~$
```

Este comando es mucho más directo y limpio que dig

Solo dice google.com has address 142.250.113.138 y ya.

```
migue31@cliente:~$ host facebook.com
facebook.com has address 157.240.25.35
facebook.com has IPv6 address 2a03:2880:f135:181:face:b00c:0:25de
facebook.com mail is handled by 10 smtpin.vvv.facebook.com.
migue31@cliente:~$
```

4. Consulta con nslookup:

```
nslookup www.ejemplo.com
```

- Observa si muestra el mismo servidor DNS y IP.

```
google.com mail is handled by 10 smtp.google.com.
migue31@cliente:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.113.139
Name:   google.com
Address: 142.250.113.138
Name:   google.com
Address: 142.250.113.113
Name:   google.com
Address: 142.250.113.102
Name:   google.com
Address: 142.250.113.101
Name:   google.com
Address: 142.250.113.100
Name:   google.com
Address: 2607:f8b0:4023:1000::8a
Name:   google.com
Address: 2607:f8b0:4023:1000::66
Name:   google.com
Address: 2607:f8b0:4023:1000::71
Name:   google.com
Address: 2607:f8b0:4023:1000::64

migue31@cliente:~$
```

```
migue31@cliente:~$ nslookup facebook.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   facebook.com
Address: 157.240.25.35
Name:   facebook.com
Address: 2a03:2880:f135:181:face:b00c:0:25de

migue31@cliente:~$ _
```

Parte 2 – Información de dominio

5. Consulta WHOIS:

```
whois ejemplo.com
```

- Identifica el registrador y la fecha de expiración.

```
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
```

```
   Domain Name: FACEBOOK.COM
   Registry Domain ID: 2320948_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.registrarsafe.com
   Registrar URL: http://www.registrarsafe.com
   Updated Date: 2025-04-23T19:08:37Z
   Creation Date: 1997-03-29T05:00:00Z
   Registry Expiry Date: 2034-03-30T04:00:00Z
   Registrar: RegistrarSafe, LLC
   Registrar IANA ID: 3237
   Registrar Abuse Contact Email: abusecomplaints@registrarsafe.com
   Registrar Abuse Contact Phone: +1-650-308-7004
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
   Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
   Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
   Name Server: A.NS.FACEBOOK.COM
   Name Server: B.NS.FACEBOOK.COM
   Name Server: C.NS.FACEBOOK.COM
   Name Server: D.NS.FACEBOOK.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-01-16T17:32:37Z <<<

For more information on Whois status codes, please visit https://icann.org/epp
```

Parte 3 – Diagnóstico y conectividad

6. Prueba de conectividad con ping:

```
ping -c 4 www.ejemplo.com
```

- Observa el tiempo de respuesta (ms).

```
migue31@cliente:~$ ping -c 4 google.com
PING google.com (142.250.72.142) 56(84) bytes of data.
64 bytes from lax17s49-in-f14.1e100.net (142.250.72.142): icmp_seq=1 ttl=63 time=77.7 ms
64 bytes from lax17s49-in-f14.1e100.net (142.250.72.142): icmp_seq=2 ttl=63 time=72.6 ms
64 bytes from lax17s49-in-f14.1e100.net (142.250.72.142): icmp_seq=3 ttl=63 time=78.7 ms
64 bytes from lax17s49-in-f14.1e100.net (142.250.72.142): icmp_seq=4 ttl=63 time=81.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 72.592/77.596/81.449/3.202 ms
migue31@cliente:~$ _
```

```
migue31@cliente:~$ ping -c 4 facebook.com
PING facebook.com (157.240.25.35) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-02-qro1.facebook.com (157.240.25.35): icmp_seq=1 ttl=63 time=21.6 ms
64 bytes from edge-star-mini-shv-02-qro1.facebook.com (157.240.25.35): icmp_seq=2 ttl=63 time=14.3 ms
64 bytes from edge-star-mini-shv-02-qro1.facebook.com (157.240.25.35): icmp_seq=3 ttl=63 time=22.5 ms
64 bytes from edge-star-mini-shv-02-qro1.facebook.com (157.240.25.35): icmp_seq=4 ttl=63 time=17.4 ms

--- facebook.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3016ms
rtt min/avg/max/mdev = 14.280/18.946/22.473/3.311 ms
```

7. Trazado de ruta con traceroute:

```
traceroute www.ejemplo.com
```

- Cuenta cuántos saltos hay hasta el destino.

```
migue31@cliente:~$ traceroute google.com
traceroute to google.com (142.250.72.142), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  1.069 ms  0.533 ms  0.525 ms
 2  172.26.160.1 (172.26.160.1)  5.437 ms  5.029 ms  3.921 ms
 3  172.17.17.5 (172.17.17.5)  3.405 ms  2.121 ms  8.678 ms
 4  172.17.17.1 (172.17.17.1)  6.628 ms  5.858 ms  5.325 ms
 5  10.10.11.20 (10.10.11.20)  4.484 ms  3.662 ms  2.764 ms
 6  201-139-182-1.blacom.net (201.139.182.1)  2.841 ms  5.944 ms  5.388 ms
 7  10.28.249.2 (10.28.249.2)  4.512 ms  3.958 ms  3.453 ms
 8  10.27.236.1 (10.27.236.1)  5.287 ms  12.183 ms  11.751 ms
 9  187-249-5-5.internetmax.maxcom.net.mx (187.249.5.5)  10.707 ms  10.214 ms  9.936 ms
10  * * *
11  * * *
12  * * *
13  201-174-244-214.transtelco.net (201.174.244.214)  12.202 ms  11.787 ms  10.973 ms
14  142.250.166.40 (142.250.166.40)  10.428 ms  21.728 ms 72.14.220.150 (72.14.220.150)  21.012 ms
15  * * 192.178.104.243 (192.178.104.243)  13.878 ms
16  142.250.212.36 (142.250.212.36)  13.480 ms  12.822 ms 142.250.210.156 (142.250.210.156)  12.470 ms
17  142.250.212.36 (142.250.212.36)  11.590 ms 192.178.104.232 (192.178.104.232)  13.421 ms *
18  192.178.110.77 (192.178.110.77)  13.508 ms 216.239.51.95 (216.239.51.95)  30.092 ms 192.178.72.237 (192.178.72.237)  12.772 ms
19  66.249.94.165 (66.249.94.165)  65.212 ms * 108.170.230.120 (108.170.230.120)  67.416 ms
20  108.170.230.120 (108.170.230.120)  67.105 ms 142.250.208.115 (142.250.208.115)  76.213 ms 74.125.243.89 (74.125.243.89)  66.178 ms
21  142.251.60.111 (142.251.60.111)  65.268 ms  64.902 ms 142.250.231.89 (142.250.231.89)  73.907 ms
22  142.251.60.111 (142.251.60.111)  65.742 ms  63.783 ms lax17s49-in-f14.1e100.net (142.250.72.142)  73.831 ms
migue31@cliente:~$
```

```
migue31@cliente:~$ traceroute facebook.com
traceroute to facebook.com (157.240.25.35), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  0.729 ms  0.380 ms  0.654 ms
 2  172.26.160.1 (172.26.160.1)  8.862 ms  8.364 ms  7.874 ms
 3  172.17.17.5 (172.17.17.5)  14.922 ms  14.601 ms  14.035 ms
 4  172.17.17.1 (172.17.17.1)  38.373 ms  95.094 ms  91.756 ms
 5  10.10.11.20 (10.10.11.20)  24.046 ms  23.538 ms  22.887 ms
 6  201-139-182-1.blacom.net (201.139.182.1)  22.348 ms  21.132 ms  20.438 ms
 7  10.28.249.2 (10.28.249.2)  19.339 ms  18.852 ms  33.709 ms
 8  10.27.236.1 (10.27.236.1)  33.015 ms  32.541 ms  31.809 ms
 9  187-249-5-5.internetmax.maxcom.net.mx (187.249.5.5)  31.085 ms  49.651 ms  48.760 ms
10  * * *
11  * * *
12  * * *
13  201-174-244-214.transtelco.net (201.174.244.214)  18.403 ms  17.720 ms  13.042 ms
14  ae1015.pr01.qro1.tfbnw.net (157.240.84.150)  12.123 ms ae1005.pr04.qro1.tfbnw.net (157.240.70.56)  11.621 ms 201-174-172-169.transtelco.net (201.174.172.16
)  10.614 ms
15  po4008.asw01.qro1.tfbnw.net (147.75.212.72)  28.243 ms  18.835 ms po4005.asw04.qro1.tfbnw.net (147.75.212.118)  23.838 ms
16  psw03.qro1.tfbnw.net (157.240.55.246)  23.326 ms psw04.qro1.tfbnw.net (157.240.55.245)  23.652 ms psw02.qro1.tfbnw.net (157.240.55.247)  23.047 ms
17  msw1ao.02.qro1.tfbnw.net (129.134.90.236)  22.578 ms msw1af.02.qro1.tfbnw.net (129.134.93.180)  21.806 ms msw1ae.02.qro1.tfbnw.net (129.134.90.246)  21.028
ms
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Parte 4 – Captura y análisis de tráfico DNS

8. Captura con tcpdump:

```
sudo tcpdump -i eth0 port 53
```

- Observa las consultas y respuestas DNS en tiempo real.
- Presiona Ctrl + C para detener.

```
migue31@cliente:~$ sudo tcpdump -i enp0s3 port 53
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

Parte 5 – Resolución paso a paso

9. Consulta recursiva con dig +trace:

```
dig +trace ejemplo.com
```

- Observa cómo se resuelve desde la raíz hasta el dominio final.

```
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> +trace google.com
;; global options: +cmd
.                       7193    IN      NS      a.root-servers.net.
.                       7193    IN      NS      m.root-servers.net.
.                       7193    IN      NS      f.root-servers.net.
.                       7193    IN      NS      c.root-servers.net.
.                       7193    IN      NS      j.root-servers.net.
.                       7193    IN      NS      l.root-servers.net.
.                       7193    IN      NS      h.root-servers.net.
.                       7193    IN      NS      k.root-servers.net.
.                       7193    IN      NS      d.root-servers.net.
.                       7193    IN      NS      i.root-servers.net.
.                       7193    IN      NS      e.root-servers.net.
.                       7193    IN      NS      b.root-servers.net.
.                       7193    IN      NS      g.root-servers.net.
;; Received 239 bytes from 127.0.0.53#53(127.0.0.53) in 0 ms

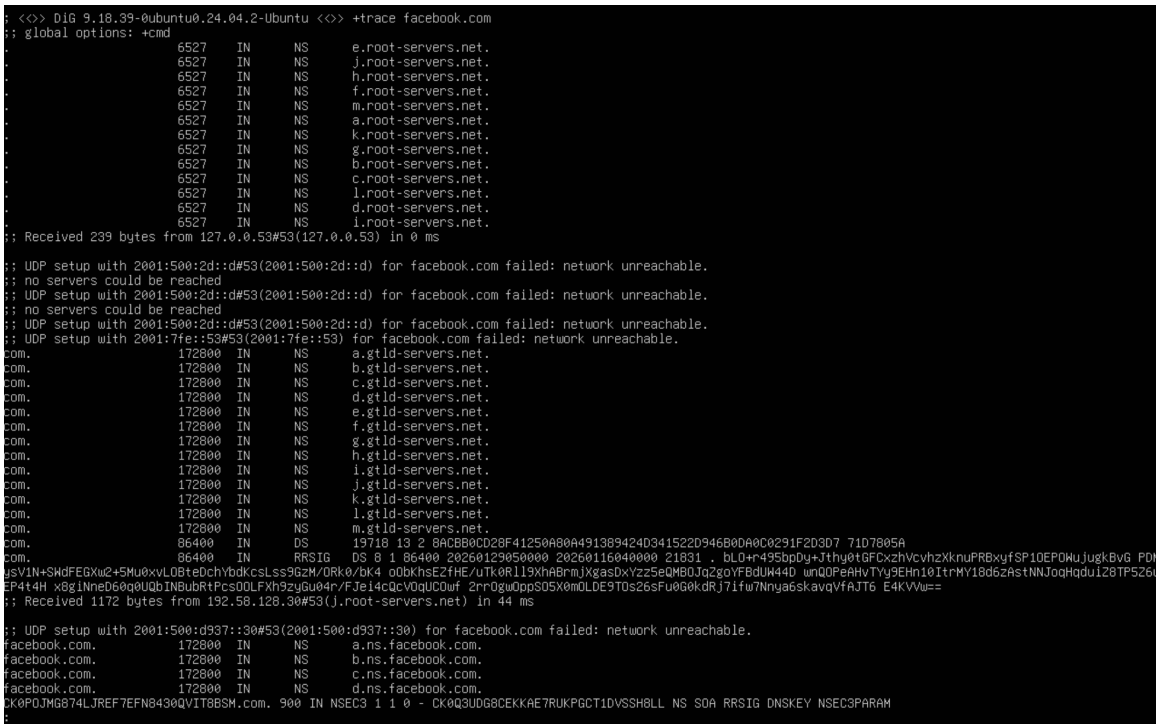
;; UDP setup with 2001:500:1::53#53(2001:500:1::53) for google.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:500:1::53#53(2001:500:1::53) for google.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:500:1::53#53(2001:500:1::53) for google.com failed: network unreachable.
com.                    172800  IN      NS      a.gtld-servers.net.
com.                    172800  IN      NS      b.gtld-servers.net.
com.                    172800  IN      NS      c.gtld-servers.net.
com.                    172800  IN      NS      d.gtld-servers.net.
com.                    172800  IN      NS      e.gtld-servers.net.
com.                    172800  IN      NS      f.gtld-servers.net.
com.                    172800  IN      NS      g.gtld-servers.net.
com.                    172800  IN      NS      h.gtld-servers.net.
com.                    172800  IN      NS      i.gtld-servers.net.
com.                    172800  IN      NS      j.gtld-servers.net.
com.                    172800  IN      NS      k.gtld-servers.net.
com.                    172800  IN      NS      l.gtld-servers.net.
com.                    172800  IN      NS      m.gtld-servers.net.
com.                    86400   IN      DS      19718 13 2 8ACBB0CD28F41250A80A491389424D341522D946B0DA0C0291F2D3D7 71D7805A
com.                    86400   IN      RRSIG   DS 8 1 86400 20260129050000 20260116040000 21831 . bLO+r495bpDy+Jthy0tGFCxzhVcvhzXknuPRBxyfSP1OEPOWujugkBvG PDM
ysV1N+SWdFEGXw2+5Mu0xvLOBteDchYbdKcsLss9GzM/ORk0/bK4 oObkhsEZfHE/uTk0Rll9XhABrmjXgasDxYzz5eQMBOJqZgoYFBdUW44D wnQOPeAHvTYy9EHn10ItrMY18d6zAstNNJoqHqdui28TP526u
EP4t4H x8giNneD60q0UQbINBubRtPcsOOLFXh9zyGu04r/FJei4cQcVOqUCOwf 2rrOgwOppSO5X0mOLDE9TOs26sFu0G0kdRj7ifw7Nnya6skavqVfAJT6 E4KVVw==
;; Received 1170 bytes from 192.58.128.30#53(j.root-servers.net) in 60 ms

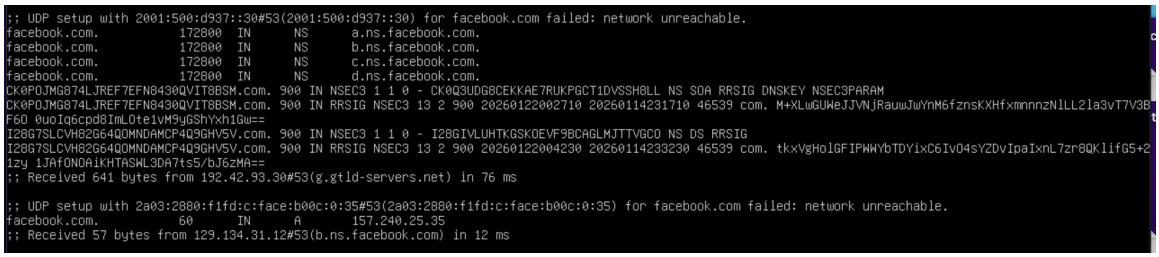
;; UDP setup with 2001:503:eea3::30#53(2001:503:eea3::30) for google.com failed: network unreachable.
;; UDP setup with 2001:503:d2d::30#53(2001:503:d2d::30) for google.com failed: network unreachable.
google.com.             172800  IN      NS      ns2.google.com.
google.com.             172800  IN      NS      ns1.google.com.
google.com.             172800  IN      NS      ns3.google.com.
google.com.             172800  IN      NS      ns4.google.com.
CK0POJMG874LJREF7EFN8430QVIT8BSM.com. 900 IN NSEC3 1 1 0 - CK0Q3UDG8CEKKAE7RUKPGCT1DVSSH8LL NS SOA RRSIG DNSKEY NSEC3PARAM
;
```

```
;; UDP setup with 2001:503:eea3::30#53(2001:503:eea3::30) for google.com failed: network unreachable.
;; UDP setup with 2001:503:d2d::30#53(2001:503:d2d::30) for google.com failed: network unreachable.
google.com.             172800  IN      NS      ns2.google.com.
google.com.             172800  IN      NS      ns1.google.com.
google.com.             172800  IN      NS      ns3.google.com.
google.com.             172800  IN      NS      ns4.google.com.
CK0POJMG874LJREF7EFN8430QVIT8BSM.com. 900 IN NSEC3 1 1 0 - CK0Q3UDG8CEKKAE7RUKPGCT1DVSSH8LL NS SOA RRSIG DNSKEY NSEC3PARAM
CK0POJMG874LJREF7EFN8430QVIT8BSM.com. 900 IN RRSIG NSEC3 13 2 900 20260122002710 20260114231710 46539 com. M+XLwGUWeJJVNjRauwJwYnM6fznsKXHfxmnnnznNlLL2la3vT7V3B
f6O 0uoIq6cpd8ImLOte1vM9yGShYxh1Gw==
S84BOR4DK28HNHPLC21BO483VOOOD5D8.com. 900 IN NSEC3 1 1 0 - S84BR9CIB2A20L3ETR1M2415ENPP99L8 NS DS RRSIG
S84BOR4DK28HNHPLC21BO483VOOOD5D8.com. 900 IN RRSIG NSEC3 13 2 900 20260123013320 20260116002320 35511 com. EEOGN1qjzEvco2KUOxzl+RjHYJLo+G2DB1EHqVrJgcQJ16OoRVSY
hkG mPBF1x2Ka5gNBl844S3M0kERWR+A9w==
;; Received 644 bytes from 192.43.172.30#53(i.gtld-servers.net) in 99 ms

google.com.             300     IN      A       192.178.56.110
;; Received 55 bytes from 216.239.32.10#53(ns1.google.com) in 44 ms

(END)
```





Preguntas para evaluar

1. ¿Cuál comando permite ver paso a paso la resolución DNS desde la raíz?

   El comando es dig con la opción +trace .

2. ¿Qué diferencia hay entre el resultado de host y dig?

   host muestra un resumen simple y directo (IPs y alias), mientras que dig muestra información técnica detallada (encabezados, tiempos de consulta, sección de respuesta, autoridad) .

3. ¿Qué comando utilizaste para identificar el registrador de un dominio?

   El comando whois .

4. ¿Qué tipo de registro DNS se utiliza para servidores de correo?

El registro **MX** (Mail Exchange) .

5. En el tcpdump, ¿qué puerto se utiliza para consultas DNS?

El puerto **53** .

6. ¿Cuántos saltos se necesitaron para llegar a www.ejemplo.com según traceroute?

   Da 22 saltos en google.com por ejemplo

7. ¿Por qué puede ser útil vaciar la caché DNS local antes de una prueba?

   Para asegurar que la prueba refleje el estado actual de la red y del servidor DNS remoto, y no una respuesta antigua guardada en tu computadora.